CEG 5303: Intelligent Autonomous Robotic Systems (IROS)

Have you wondered about building a robot that could navigate and manipulate objects in your environment?

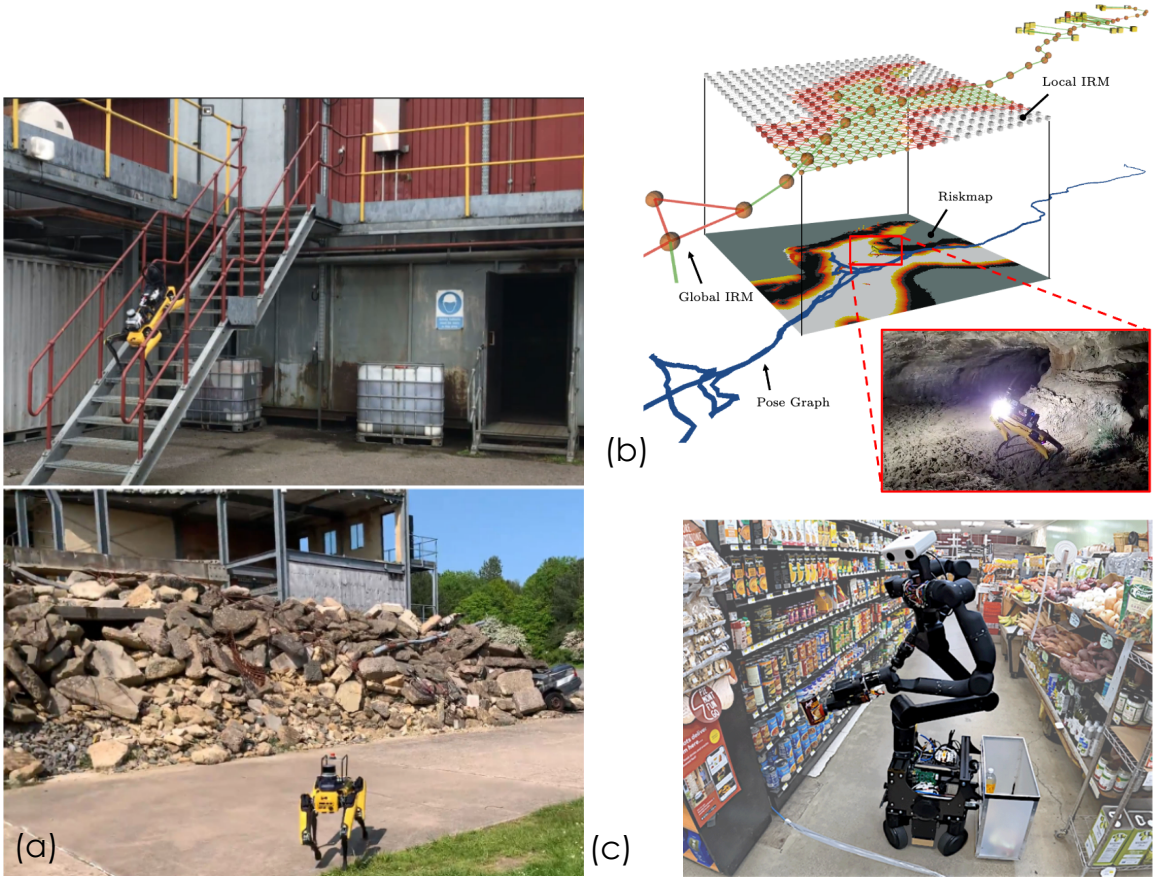

Figure: (a) AutoInspect: A long-term autonomous inspection and monitoring system that operates a robot over several days with minimal intervention. [Link]. (b) NeBula @ DARPA SubT Challenge. Multi-Robot Autonomous Mapping and Exploration in Extreme Environments. [Link]. (c) Toyota Research Institute's Mobile Manipulator Robot for Autonomous Shopping. [Link].

This course offers a systematic description of how to design and implement the core components of such an intelligent robotic system. You'll start by developing the essential mathematical foundations, including 3D geometry and Lie groups, then implement core perception modules through camera calibration, visual-inertial odometry, and mapping algorithms that enable robots to understand their environment. You'll explore cutting-edge deep learning approaches, including foundation models (CLIP, SAM) and work with perception pipelines that give robots human-level visual understanding. The course covers navigation modules with terrain estimation for mobile platforms and grasping modules using modern end-to-end deep learning techniques for manipulation tasks. Each module is taught through targeted implementation exercises using open-source repositories, giving you practical experience with key robotic algorithms and their real-world application. Through this modular approach with hands-on coding exercises, you'll gain a deep understanding of how each component works individually and how they would

integrate in a complete mobile manipulation system. This course will provide you with the requisite foundational knowledge and practical implementation skills needed to tackle complex autonomous robotics challenges in research and industry.

Teaching Staff

Rajat Talak, Assistant Professor
Electrical and Computer Engineering Department

Course Prerequisites

The course has no prerequisite NUS courses. However, background in Linear Algebra, Calculus, and computational skills using Python, PyTorch, and C++ would be helpful.

Lecture Time & Venue

Monday, 6 pm – 9 pm. UTSRC LT50: https://nusmods.com/venues/UTSRC-LT50
The course requires in-person attendance.

Lectures and Schedule

Date	Lecture Topic	Note
Jan 12, 2026	Introduction and Overview	
Jan 19, 2026	3D Geometry Basics I	
Jan 26, 2026	3D Geometry Basics II	
Feb 2, 2026	Vision Basics I	
Feb 9, 2026	Vision Basics II	
Feb 16, 2026		Lecture Cancelled
Feb 23, 2026		Reading Week
Mar 2, 2026		Mid-Term Exam
Mar 9, 2026	Localization	
Mar 16, 2026	Mapping	
Mar 23, 2026	Place Recognition	
Mar 30, 2026	Navigation	
Apr 6, 2026	Grasping and Manipulation	
Apr 13, 2026	Sensors and Case Studies	
Apr 20, 2026		Reading Week
Apr 27, 2026		Final Exam: TBD
May 4, 2026		Final Exam: TBD

Assignments and Grading

The course will conduct three take-home assignments, a mid-term test, and a final exam. The take-home assignments will be a set of questions that will have to be answered and handed in by the deadline. The mid-term and final exam will be in-person exams and will be a maximum of three-hour duration. Three assignments will contribute to 30% of your final score. The mid-term and final exam will contribute to 30% and 40% of your final score, respectively.

Late Work Policy

Late work on the Assignments will receive reduced credit according to the following policy: if the grade achieved by a submission is X, the grade resulting from a late submission is computed as: X_delayed = X * max(1 - 0.15 * d, 0), where d is the number of days of delay. For instance, if X = 10 and there is no late submission, then X_delayed = 10, but if the assignment is submitted 2 days after the deadline, then X_delayed = 10 * (1 - 0.3) = 7.

Contact

If you have any questions, please reach out to Rajat Talak (talak@nus.edu.sg)